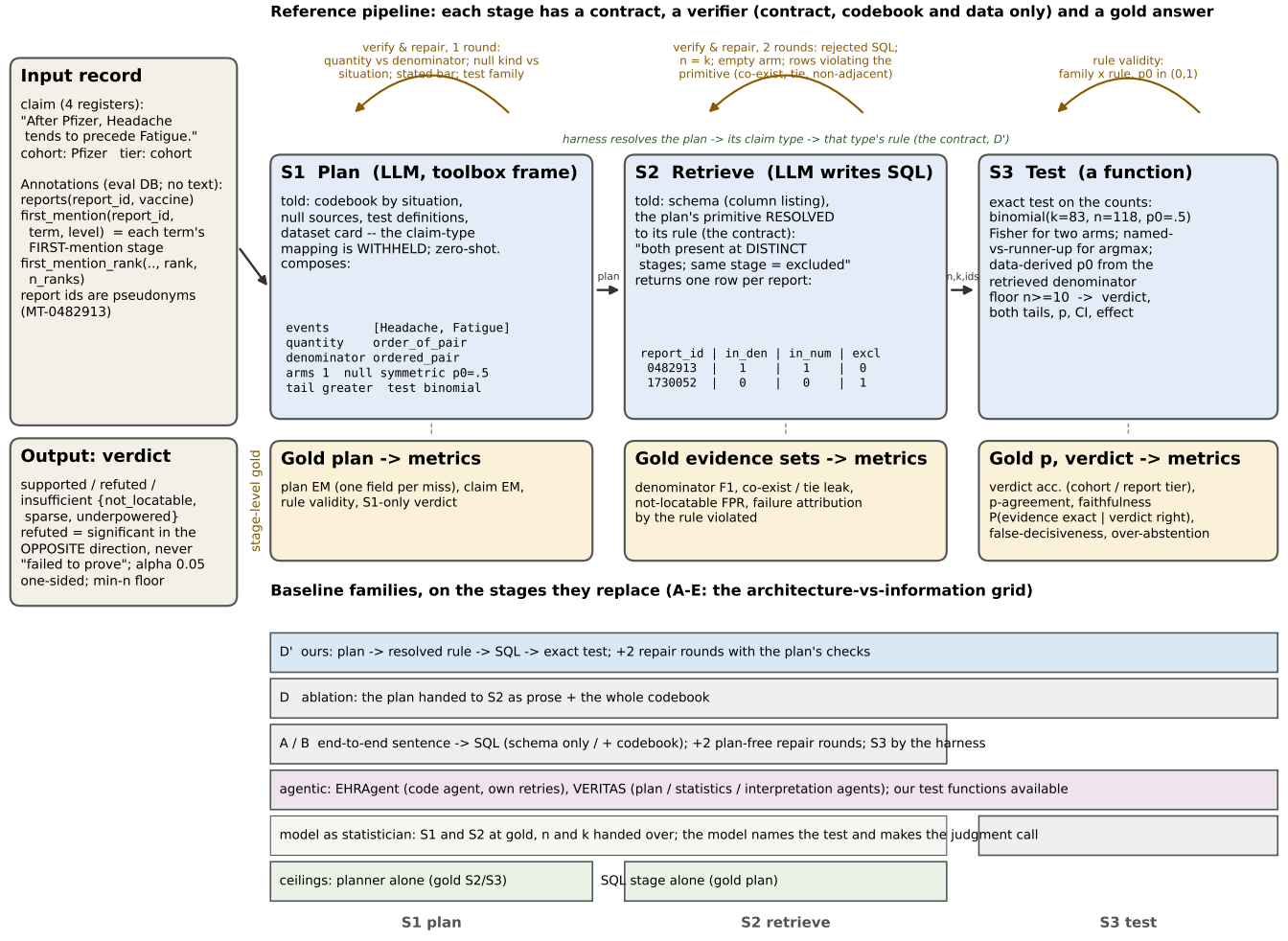

Reference pipeline: each stage has a contract, a verifier (contract, codebook and data only) and a gold answer
verify & repair, 1 round: quantity vs denominator; null kind vs situation; stated bar; test family
verify & repair, 2 rounds: rejected SQL; n = k; empty arm; rows violating the primitive (co-exist, tie, non-adjacent)
rule validity: family x rule, p0 in (0,1)
harness resolves the plan -> its claim type -> that type's rule (the contract, D')
Input record
claim (4 registers):
"After Pfizer, Headache tends to precede Fatigue."
cohort: Pfizer tier: cohort
Annotations (eval DB; no text):
reports(report_id, vaccine)
first_mention(report_id, term, level) = each term's FIRST-mention stage
first_mention_rank(.., rank, n_ranks)
report ids are pseudonyms (MT-0482913)
S1 Plan (LLM, toolbox frame)
told: codebook by situation, null sources, test definitions, dataset card -- the claim-type mapping is WITHHELD; zero-shot. composes:
events [Headache, Fatigue]
quantity order_of_pair
denominator ordered_pair
arms 1 null symmetric p0=.5
tail greater test binomial
plan
S2 Retrieve (LLM writes SQL)
told: schema (column listing), the plan's primitive RESOLVED to its rule (the contract): "both present at DISTINCT stages; same stage = excluded" returns one row per report:
report_id | in_den | in_num | excl
0482913 | 1 | 1 | 0
1730052 | 0 | 0 | 1
n,k,ids
S3 Test (a function)
exact test on the counts: binomial(k=83, n=118, p0=.5) Fisher for two arms; named-vs-runner-up for argmax; data-derived p0 from the retrieved denominator floor n>=10 -> verdict, both tails, p, CI, effect
Output: verdict
supported / refuted / insufficient {not_locatable, sparse, underpowered}
refuted = significant in the OPPOSITE direction, never "failed to prove"; alpha 0.05 one-sided; min-n floor
stage-level gold
Gold plan -> metrics
plan EM (one field per miss), claim EM, rule validity, S1-only verdict
Gold evidence sets -> metrics
denominator F1, co-exist / tie leak, not-locatable FPR, failure attribution by the rule violated
Gold p, verdict -> metrics
verdict acc. (cohort / report tier), p-agreement, faithfulness P(evidence exact | verdict right), false-decisiveness, over-abstention
Baseline families, on the stages they replace (A-E: the architecture-vs-information grid)
D' ours: plan -> resolved rule -> SQL -> exact test; +2 repair rounds with the plan's checks
D ablation: the plan handed to S2 as prose + the whole codebook
A / B end-to-end sentence -> SQL (schema only / + codebook); +2 plan-free repair rounds; S3 by the harness
agentic: EHRAgent (code agent, own retries), VERITAS (plan / statistics / interpretation agents); our test functions available
model as statistician: S1 and S2 at gold, n and k handed over; the model names the test and makes the judgment call
ceilings: planner alone (gold S2/S3)
SQL stage alone (gold plan)
S1 plan
S2 retrieve
S3 test